

Question

The following chemical processes are described:

1. In early industry, sodium amide reacts with carbon to produce a highly toxic substance.
2. Orpiment reacts with a hydrochloric acid solution of stannous chloride to yield a substance that can be used as an insecticide or antidote.
3. When excess sodium cyanide is added to a mercury-containing container, the mercury completely dissolves after some time.
4. Dry $\text{NI}_3 \cdot \text{NH}_3$ is highly prone to explosive decomposition, producing two iodine-containing substances.
5. Dry iron powder reacts with excess nitric acid of varying concentrations, yielding products that may be **A**, **B**, **C**, **D**, **E**, with nitrogen mass percentages of approximately 35%, 100%, 64%, 47%, 31%, respectively.

Write and balance the chemical equations for these processes (with coefficients in the simplest integer ratio).

Which of the following statements is correct?

- A.** The equation for Process 1 does not generate gas.
- B.** The sum of the reactant coefficients in the equation for Process 2 is 7.
- C.** The sum of the reactant coefficients in the equation for Process 3 is 12.
- D.** The sum of the product coefficients in the equation of process 4 is not a multiple of 4.
- E.** Process 5 generates the equation for **A**, where the sum of product coefficients is the smallest among the equations for generating **A-E**.
- F.** Process 5 generates the equation for **B**, where the sum of product coefficients is the largest among the equations generating **A-E**.

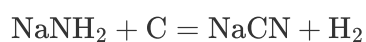
- G.** Process 5 generates the equation for **C**, where the sum of reactant coefficients is the largest among all equations producing **A-E**, without exception.
- H.** Process 5 generates the equation for **D**, where the sum of reactant coefficients is the smallest among the equations producing **A-E**.
- I.** Process 5 generates the equation for **E**, where the sum of the product coefficients is the only odd number among the equations for **A-E**.
- J.** None of the above options are correct

Answer

Correct Answer: **H**

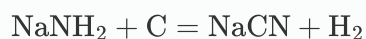
Detailed Explanation

Process 1: The highly toxic substance is clearly sodium cyanide, while hydrogen gas is released. Option A is incorrect. The equation is

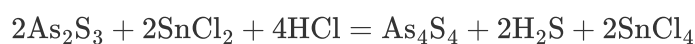


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Process 2: For insecticidal detoxification, the substance is clearly realgar As_4S_4 . Divalent tin is used for reduction and is itself oxidized to tetravalent tin. The equation is



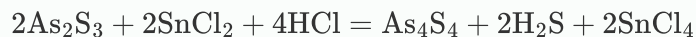
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The insecticidal detoxification substance is realgar As_4S_4

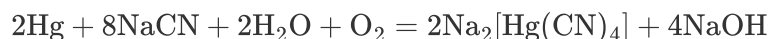
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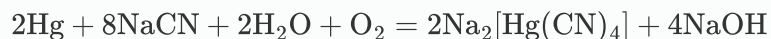
The sum of the reactant coefficients is 8, so option B is incorrect. If considering the form of Sn(II) in the hydrochloric acid system as H_2SnCl_4 and Sn(IV) as H_2SnCl_6 , the sum of the equation coefficients still does not equal 7.

Process 3: Divalent mercury can be coordinated by cyanide to form the stable $\text{Hg}(\text{CN})_4^{2-}$. During the dissolution of mercury, it is oxidized by oxygen in the air. Thus, the chemical equation is written as



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The sum of the reactant coefficients is 13, so option C is incorrect.

Process 4: $\text{NI}_3 \cdot \text{NH}_3$ undergoes an explosion because the process is a highly intense redox reaction, producing stable species. It can be deduced that nitrogen gas and iodine are formed, while the coordinated ammonia combines with the iodide ions generated from disproportionation. Thus, the equation is:



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The sum of the product coefficients is 20, which is a multiple of 4, so option D is incorrect.

Process 5: First, deduce **A-E**. Clearly, **B** is nitrogen gas. Then, assume the general formula for **A-E** is $N_xH_yO_z$ and infer based on mass fractions. The final results for **A-E** are ammonium nitrate, nitrogen gas, nitrous oxide, nitric oxide, and nitrogen dioxide.

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A-E are ammonium nitrate, nitrogen gas, nitrous oxide, nitric oxide, and nitrogen dioxide

Since it is a reaction between iron powder and nitric acid, the product must be iron(III) nitrate. Thus, the following five redox equations are balanced:

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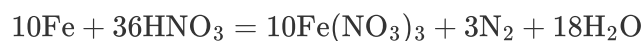
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The product must be iron(III) nitrate



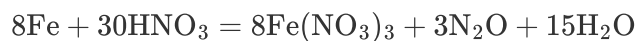
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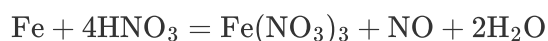
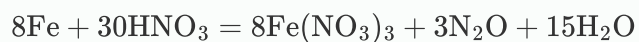
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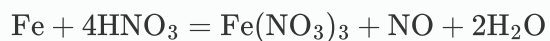
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Based on the sum of the product and reactant coefficients, only option H is correct among options E-I.

In conclusion, the answer is H.